ENVIRONMENTAL PRODUCT DECLARATION

Limestone Aggregates - Latomeia Prevezis AEVE

In accordance with: ISO 14025:2006

Products included in the EPD:

Sand 0/4, Fine gravel (garbili) 4/11.2, Gravel (haliki) 16/31.5, 3A 0/22.4

An EPD may be updated or depublished if conditions change. To find the latest version of the EPD and to confirm its validity, see www.environdec.com

EPD of multiple products based on the average results of the product group

EPD Owner
LATOMEIA PREVEZIS A.E.V.E.

Programme
International EPD System
www.environdec.com

Programme operator
EPD International AB

Registration number
EPD-IES-0025910:004

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2025-10-14

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2030-10-05

EPD INTERNATIONAL EPD SYSTEM

ECO PLATFORM EPD VERIFIED

ΛΑΤΟΜΕΙΑ ΠΡΕΒΕΖΗΣ Α.Ε.Β.Ε.

GENERAL INFORMATION

Programme information

Programme	International EPD System
Address	EPD International AB Box 210 60 SE-100 31 Stockholm Sweden
Website	www.environdec.com
E-mail	support@environdec.com

Product category rules

CEN standard EN 15804 serves as the Core Product Category Rules (PCR)	
Product Category Rules (PCR)	2019:14 Construction products (EN 15804+A2) (version 2.0.1) 2.0.1
PCR review was conducted by	The Technical Committee of the International EPD System. See www.environdec.com for a list of members. Review chair: Rob Rouwette (chair), Noa Meron (co-chair). The review panel may be contacted via the Secretariat www.environdec.com/support.

Verification

LCA accountability	Dionysis Giakoumelos, support@dgiakoumelos.com, LATOMEIA PREVEZIS A.E.V.E.
Independent third-party verification of the declaration and data, according to ISO 14025:2006, via	☑ EPD verification through an individual EPD verification ☐ EPD verification through EPD Process Certification* ☐ EPD verification through a fully pre-verified tool
Third-party verifier	Business Quality Verification P.C.
Accredited by	Hellenic Accreditation System ESYD
Accredited certification body address	Greece
Procedure for follow-up of data during EPD validity involves third party verifier	☑ Yes ☐ No
*EPD Process Certification involves an accredited certification body certifying and periodically auditing the EPD process and conducting external and independent verification of EPDs that are regularly published. More information can be found in the General Programme Instructions on www.environdec.com. International EPD System.	

Ownership and limitations on use of EPD

Limitations

EPDs within the same product category but published in different EPD programmes, may not be comparable. For two EPDs to be comparable, they shall be based on the same PCR (including the same first-digit version number) or be based on fully aligned PCRs or versions of PCRs; cover products with identical functions, technical performances and use (e.g. identical declared/functional units); have identical scope in terms of included life-cycle stages (unless the excluded life-cycle stage is demonstrated to be insignificant); apply identical impact assessment methods (including the same version of characterisation factors); and be valid at the time of comparison.

Ownership

The EPD Owner has the sole ownership, liability, and responsibility for the EPD.

INFORMATION ABOUT EPD OWNER

EPD Owner	LATOMEIA PREVEZIS A.E.V.E.
Contact person name	Markos Nastas
Contact person e-mail	nastasmarkos@hotmail.com
Organisation address	Greece PREVEZA 48300 Thesi «Boufos», Thesprotiko Prevezis

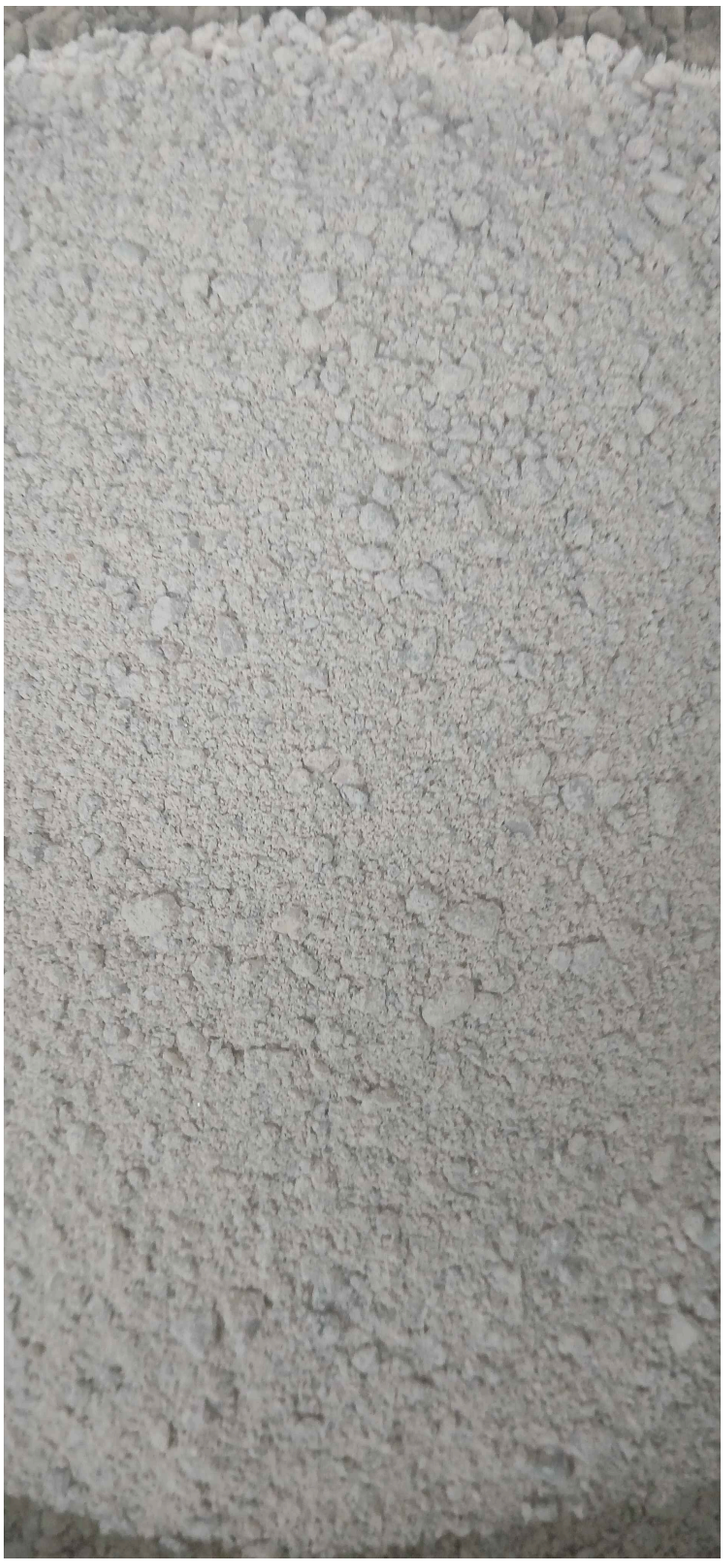
Description of the organisation of the EPD Owner

PREVEZIS QUARRIES S.A. was founded in 1996 and is a member company of the ACHERON CONCRETE QUARRIES S.A. group.
Understanding the durable link between the natural wealth of the region and the contracting-construction industry, it provides reliable materials produced based on certified processes.
The company supplies the domestic market with the highest quality bulk aggregates for use in big technical public and private projects and constructions. In addition, it offers secondary solutions, such as certified quality readymix concrete, as well as circular economy solutions through sustainable management.
It takes care of the quality of each of our collaborations, creating honest relationships, for the benefit of each of its customers projects and the successful completion of the most ambitious construction projects. It operates responsibly with quality assurance of production processes, environmental management certification and good practices according to European standards.

PRODUCT INFORMATION

Product name	Sand 0/4
Product identification	Crushed limestone sand 0/4 mm from primary crushing, CE-marked according to EN 12620:2002+A1:2008 and EN 13043:2002+AC:2004
Product description	The product, commercially referred to as "Sand 0/4", is a natural crushed limestone sand with a particle size of 0/4 mm, produced through primary crushing of limestone rock at a single quarry. It is CE-marked and complies with EN 12620 for concrete and EN 13043 for bituminous mixtures and surface treatments. The material is suitable for a wide range of applications, including infrastructure and roadworks.
Technical purpose of product	The product is intended for use as a construction aggregate, primarily as a fine aggregate (0/4 mm) in concrete production and bituminous mixtures. It meets the technical requirements specified in the relevant European standards for aggregates used in these sectors.
Manufacturing or service provision description	The production process begins with blasting of natural limestone rock using explosives. The blasted material is transported by diesel-powered mobile equipment (trucks) to a crusher, where it is crushed to the desired size. Environmental management measures are in place to minimize dust, noise, and water impacts.
Material properties	Volumetric mass density: 1650 kg/m^3
Manufacturing site	Latomeia Prevezis AEVE Greece "Boufos" area, Thesprotiko, Preveza, Greece 483 00
UN CPC code	15320. Pebbles, gravel, broken or crushed stone, macadam; granules, chippings and powder of stone
Geographical scope(s)	Greece

PRODUCT IMAGES



PRODUCT INFORMATION

Product name	Fine gravel (garbili) 4/11.2
Product identification	Crushed limestone fine gravel (garbili) 4/11.2 mm from primary crushing, CE-marked according to EN 12620:2002+A1:2008 and EN 13043:2002+AC:2004
Product description	The product, commercially referred to as "Garbili 4/11.2", is a natural crushed limestone with a particle size of 4/11.2 mm, produced through primary crushing of limestone rock at a single quarry. It is CE-marked and complies with EN 12620 for concrete and EN 13043 for bituminous mixtures and surface treatments. The material is suitable for a wide range of applications, including infrastructure and roadworks.
Technical purpose of product	The product is intended for use as a construction aggregate, primarily as garbili 4/11.2 mm in concrete production and bituminous mixtures. It meets the technical requirements specified in the relevant European standards for aggregates used in these sectors.
Manufacturing or service provision description	The production process begins with blasting of natural limestone rock using explosives. The blasted material is transported by diesel-powered mobile equipment (trucks) to a crusher, where it is crushed to the desired size. Environmental management measures are in place to minimize dust, noise, and water impacts.
Material properties	Volumetric mass density: 1650 kg/m^3
Manufacturing site	Latomeia Prevezis AEVE Greece "Boufos" area, Thesprotiko, Preveza, Greece 483 00
UN CPC code	15320. Pebbles, gravel, broken or crushed stone, macadam; granules, chippings and powder of stone
Geographical scope(s)	Greece

PRODUCT IMAGES

PRODUCT INFORMATION

Product name	Gravel (haliki) 16/31.5
Product identification	Crushed limestone gravel (haliki) 16/31.5 mm from primary crushing, CE-marked according to EN 12620:2002+A1:2008 and EN 13043:2002+AC:2004.
Product description	The product, commercially referred to as "Haliki 16/31.5", is a natural crushed limestone with a particle size of 16/31.5 mm, produced through primary crushing of limestone rock at a single quarry. It is CE-marked and complies with EN 12620 for concrete and EN 13043 for bituminous mixtures and surface treatments. The material is suitable for a wide range of applications, including infrastructure and roadworks.
Technical purpose of product	The product is intended for use as a construction aggregate, primarily as haliki 16/31.5 mm in concrete production and bituminous mixtures. It meets the technical requirements specified in the relevant European standards for aggregates used in these sectors.
Manufacturing or service provision description	The production process begins with blasting of natural limestone rock using explosives. The blasted material is transported by diesel-powered mobile equipment (trucks) to a crusher, where it is crushed to the desired size. Environmental management measures are in place to minimize dust, noise, and water impacts.
Material properties	Volumetric mass density: 1650 kg/m^3
Manufacturing site	Latomeia Prevezis AEVE Greece "Boufos" area, Thesprotiko, Preveza, Greece 483 00
UN CPC code	15320. Pebbles, gravel, broken or crushed stone, macadam; granules, chippings and powder of stone
Geographical scope(s)	Greece

PRODUCT IMAGES

PRODUCT INFORMATION

Product name	3A 0/22.4
Product identification	Crushed limestone aggregate 0/22.4 mm (3A), produced by blending sand, fine gravel and gravel (haliki) from primary crushing, CE-marked according to EN 13242:2002+A1:2007
Product description	The product, commercially referred to as “3A 0/22.4”, is a natural crushed limestone aggregate with a particle size of 0/22.4 mm, produced by blending sand, fine gravel and gravel (haliki) stone obtained from primary crushing of limestone rock at a single quarry. It is CE-marked and complies with EN 13242 for unbound and hydraulically bound materials for use in civil engineering work and road construction. The material is suitable for a wide range of applications, including infrastructure projects and roadworks.
Technical purpose of product	The product is intended for use as a construction aggregate, primarily as an unbound graded aggregate (0/22.4 mm) for road construction and civil engineering applications, such as base and sub-base layers, embankments and general infrastructure works. It meets the technical requirements specified in EN 13242 for aggregates used in unbound and hydraulically bound materials.
Manufacturing or service provision description	The production process begins with blasting of natural limestone rock using explosives. The blasted material is transported by diesel-powered mobile equipment (trucks) to a crusher, where it is crushed to the desired size. For this product, different aggregate fractions are blended to achieve the required particle size distribution of 0/22.4 mm. Environmental management measures are in place to minimize dust, noise, and water impacts.
Material properties	Volumetric mass density: 1650 kg/m^3
Manufacturing site	Latomeia Prevezis AEVE Greece "Boufos" area, Thesprotiko, Preveza, Greece 483 00
UN CPC code	15320. Pebbles, gravel, broken or crushed stone, macadam; granules, chippings and powder of stone
Geographical scope(s)	Greece

PRODUCT IMAGES

CONTENT DECLARATION

Content declaration of multiple products	The content declaration represents what material all concerned products in this EPD consist of, as all products have the same origin (blasted limestone). Therefore, the product material is the same for all products besides from the sorting size.
Hazardous and toxic substances	The product does not contain any substances from the SVHC candidate list in concentrations exceeding 0.1% of its weight.

PRODUCT CONTENT

Content name	Mass, kg	Post-consumer recycled material, mass-% of product	Biogenic material, mass-% of product	Biogenic material[1], kg C/declared unit
Limestone	1000	0	0	0
Total	**1000**	**0**	**0**	**0**
Note 1	1 kg biogenic carbon is equivalent to 44/12 kg of CO_2			

LCA INFORMATION

EPD based on declared or functional unit	Declared unit
Declared unit and reference flow	Aggregates Mass: 1000 kg
Conversion factor to mass	1
Are infrastructure or capital goods included in any upstream, core or downstream processes?	☐ Yes ☑ No
Data sources used for this EPD	ecoinvent database (general) ecoinvent 3.9.1 database
LCA Software	OpenLCA OpenLCA 2.4.0
Version of the EN 15804 reference package	EF Reference Package 3.1
Characterisation methods	The characterisation method applied in the life cycle impact assessment (LCIA) is the Environmental Footprint (EF) method, version 3.1. The method was implemented using OpenLCA v2.0.4 with the ecoinvent v3.9.1 database.
Technology description including background system	The foreground system includes typical quarrying operations for limestone aggregates, including blasting with explosives, transport of raw material using diesel-powered mobile equipment, crushing in a stationary crusher, and screening to obtain the desired fractions. The background system includes data for diesel production and combustion, explosives production and other upstream processes such as equipment use. These data were sourced from the ecoinvent v3.9.1 database and modelled using OpenLCA v2.0.4.
Scrap (recycled material) inputs contribution level	Less than 10% of the GWP-GHG results in modules A1-A3 come from scrap inputs

Data quality assessment

Description of data quality assessment and reference years	The Life Cycle Assessment (LCA) for this Environmental Product Declaration is based on both primary and secondary data. Primary data were collected directly from the manufacturer's production processes and site specific operations for the reference year 2024. The quality of the relevant data used for the EPD in terms of its time, geography and technology representativeness using EN 15804:2012+A2:2019, Annex E. The relevant data assessed included no "poor" or "very poor" data.

DATA QUALITY ASSESSMENT

Process name	Source type	Source	Reference year	Data category	Share of primary data, of GWP-GHG results for A1-A3
Diesel burned in building machine	Database	Ecoinvent v.3.9.1	2022	Secondary data	
Generation of electricity used in manufacturing of product	Collected data	DAPEEP SA	2024	Primary data	39.28%
Explosives production	Database	Ecoinvent v.3.9.1	2022	Secondary data	
Diesel production	Database	Ecoinvent v.3.9.1	2022	Secondary data	
Transportation	Database	Ecoinvent v.3.9.1	2022	Primary data	0.51%
Other processes	Database	Ecoinvent v.3.9.1	2022	Secondary data	
Total share of primary data, of GWP-GHG results for A1-A3					**39.79%**
Note	The share of primary data is calculated based on GWP-GHG results. It is a simplified indicator for data quality that supports the use of more primary data to increase the representativeness of and comparability between EPDs. Note that the indicator does not capture all relevant aspects of data quality and is not comparable across product categories.				

ELECTRICITY USED IN THE MANUFACTURING PROCESS IN A3 (A5 FOR SERVICES)

Type of electricity mix	Residual electricity mix on the market	
Energy sources	Hydro	1.74%
	Wind	21.58%
	Solar	17.93%
	Biomass	1.13%
	Geothermal	0.03%
	Waste	0%

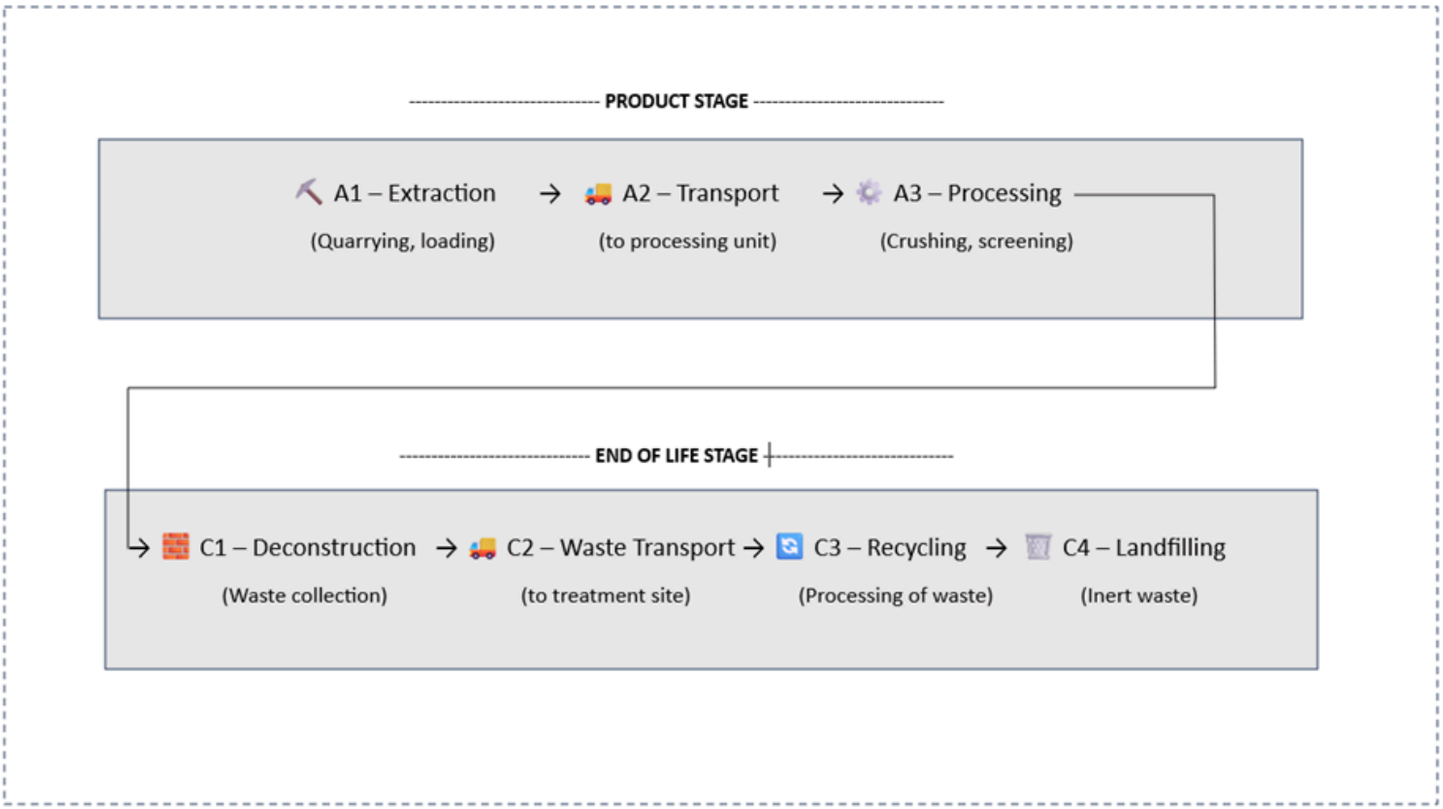
	Nuclear	0%
	Natural gas	39.39%
	Coal	6.28%
	Oil	7.49%
	Peat	0%
	Other	4.52%
Climate impact (GWP-GHG):	0.56 kg CO_2 eq./kWh	

SYSTEM BOUNDARY

Description of the System boundary	a) Cradle to gate with modules C1-C4 and module D (A1-A3 + C + D).
Excluded modules	No, there is no excluded module, or there are no excluded modules

	Product stage			Construction process stage		Use stage							End of life stage				Beyond product life cycle
	Raw material supply	Transport	Manufacturing	Transport to site	Construction installation	Use	Maintenance	Repair	Replacement	Refurbishment	Operational energy use	Operational water use	De-construction demolition	Transport	Waste processing	Disposal	Reuse-Recovery-Recycling-potential
Module	A1	A2	A3	A4	A5	B1	B2	B3	B4	B5	B6	B7	C1	C2	C3	C4	D
Modules declared	X	X	X	ND	ND	ND	ND	ND	ND	ND	ND	ND	X	X	X	X	X
Geography	Greece	Greece	Greece	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	Greece	Greece	Greece	Greece	Greece
Share of specific data	39.79%			-	-	-	-	-	-	-	-	-	-	-	-	-	-
Variation - products	0%			-	-	-	-	-	-	-	-	-	-	-	-	-	-
Variation - sites	0%			-	-	-	-	-	-	-	-	-	-	-	-	-	-
Disclaimer	The share of specific/primary data and both variations (products and sites) refer to GWP-GHG results only.																

Process flow diagram(s) related images



DEFAULT SCENARIO

Name of the default scenario	Recycling and Disposal Scenario for Construction Aggregates
Description of the default scenario	The product is assumed to be used primarily in concrete production. At the end of the building's life, the construction and demolition waste (CDW) containing the product is transported to a recycling facility. It is considered that 50% of the material will be recycled and reused as secondary aggregates, while the remaining 50% will be disposed of as waste. This default end-of-life scenario reflects common regional practices and current waste management regulations.

Module C: End-of-life

Explanatory name of the default scenario in module C	Recycling and Disposal
Description of the default scenario in module C	The product is assumed to be used primarily in concrete production. At the end of the building's life, the construction and demolition waste (CDW) containing the product is transported to a recycling facility. It is considered that 50% of the material will be recycled and reused as secondary aggregates, while the remaining 50% will be disposed of as waste. This default end-of-life scenario reflects common regional practices and current waste management regulations

Module C information	Value	Unit
Collection process, collected separately	0	kg
Collection process, collected with mixed construction waste	1000	kg
Recovery system, for re-use	0	kg
Recovery system, for recycling	500	kg
Recovery system, energy recovery	0	kg
Disposal, for final deposition	500	kg
Waste transport, distance	20	km
Vehicle type	lorry 16-32 metric ton, EURO5	
Diesel consumption for a building demolition	70	MJ

Module D: Beyond product life cycle

Explanatory name of the default scenario in module D	Secondary use
Description of the default scenario in module D	At the end of the product's life, construction and demolition waste containing the aggregates is collected and transported to recycling facilities. Approximately 50% of the material is recycled and reused as secondary aggregates, substituting virgin limestone in new construction and civil engineering projects. This recycling reduces the demand for primary raw materials and minimizes environmental impacts associated with extraction

ADDITIONAL SCENARIO 1

Name of the additional scenario	Alternative EOL Full Reuse Scenario
Description of the additional scenario	In this alternative end-of-life scenario, it is assumed that 100% of the product is recovered from construction and demolition waste CDW) and reused as secondary aggregates in new construction applications. No material is sent to landfill or incineration. The material is transported to a recycling facility and reused without significant processing. It represents an idealized best-case situation and does not reflect current averag

Module C: End-of-life

Description of the additional scenario in module C	In this alternative end-of-life scenario, it is assumed that 100% of the product is recovered from construction and demolition waste CDW) and reused as secondary aggregates in new construction applications. No material is sent to landfill or incineration. The material is transported to a recycling facility and reused without significant processing. It represents an idealized best-case situation and does not reflect current averag

Module C information	Value	Unit
Collection process, collected separately	0	kg
Collection process, collected with mixed construction waste	1000	kg
Recovery system, for re-use	0	kg
Recovery system, for recycling	1000	kg
Recovery system, energy recovery	0	kg
Disposal, for final deposition	0	kg
Waste transport, distance	20	km
Vehicle type	lorry 16-32 metric ton, EURO5	
Diesel consumption for a building demolition	70	MJ

Module D: Beyond product life cycle

Description of the additional scenario in module D	In this scenario, 100% of the product is reused as recycled aggregate, substituting an equivalent amount of primary aggregate.

ADDITIONAL SCENARIO 2

Name of the additional scenario	Alternative EOL Full Landfilling Scenario
Description of the additional scenario	In this alternative end-of-life scenario, it is assumed that 100% of the product contained in construction and demolition waste CDW) is disposed of in landfill, with no reuse or recycling. This represents a worst-case scenario for end-of-life handling. The material is collected and transported to an inert landfill site without any further treatment.

Module C: End-of-life

Description of the additional scenario in module C	In this alternative end-of-life scenario, it is assumed that 100% of the product contained in construction and demolition waste CDW) is disposed of in landfill, with no reuse or recycling. This represents a worst-case scenario for end-of-life handling. The material is collected and transported to an inert landfill site without any further treatment.

Module C information	Value	Unit
Collection process, collected separately	0	kg
Collection process, collected with mixed construction waste	1000	kg
Recovery system, for re-use	0	kg
Recovery system, for recycling	0	kg
Recovery system, energy recovery	0	kg
Disposal, for final deposition	1000	kg
Waste transport, distance	20	km
Vehicle type	lorry 16-32 metric ton, EURO5	
Diesel consumption for a building demolition	70	MJ

Module D: Beyond product life cycle

Description of the additional scenario in module D	In this scenario, 100% of the product is disposed of in landfill. No material is recovered, reused, or recycled. Therefore, no avoided impacts are reported in Module D.

ENVIRONMENTAL PERFORMANCE

The estimated impact results are only relative statements, which do not indicate the endpoints of the impact categories, exceeding threshold values, safety margins and/or risks.

Mandatory environmental performance indicators according to EN 15804

Impact category	Indicator	Unit	A1-A3	A4	A5	B1	B2	B3	B4	B5	B6	B7	C1	C2	C3	C4	D
Climate change - total	GWP-total	kg CO_2 eq.	2.57E+0	ND	ND	ND	ND	ND	ND	ND	ND	ND	6.95E+0	3.85E+0	2.17E+0	3.04E+0	-5.70E-1
Climate change - fossil	GWP-fossil	kg CO_2 eq.	2.57E+0	ND	ND	ND	ND	ND	ND	ND	ND	ND	9.16E+1	5.45E+1	2.86E+1	7.63E+1	-5.68E-1
Climate change - biogenic	GWP-biogenic	kg CO_2 eq.	2.56E-3	ND	ND	ND	ND	ND	ND	ND	ND	ND	8.71E-4	9.98E-4	2.76E-4	1.20E-3	-1.81E-3
Climate change - land use and land-use change	GWP-luluc	kg CO_2 eq.	3.80E-4	ND	ND	ND	ND	ND	ND	ND	ND	ND	7.81E-4	1.98E-3	2.44E-4	1.83E-3	-1.37E-4
Ozone depletion	ODP	kg CFC-11 eq.	9.13E-8	ND	ND	ND	ND	ND	ND	ND	ND	ND	1.10E-7	5.75E-8	3.45E-8	8.80E-8	-5.91E-8
Acidification	AP	mol H^+ eq.	1.63E-2	ND	ND	ND	ND	ND	ND	ND	ND	ND	6.44E-2	1.36E-2	2.01E-2	2.29E-2	1.00E-3
Eutrophication aquatic freshwater	EP-freshwater	kg P eq.	6.66E-4	ND	ND	ND	ND	ND	ND	ND	ND	ND	2.13E-4	3.12E-4	6.66E-5	2.53E-4	-4.72E-4
Eutrophication aquatic marine	EP-marine	kg N eq.	5.60E-3	ND	ND	ND	ND	ND	ND	ND	ND	ND	2.98E-2	4.47E-3	9.31E-3	8.79E-3	1.99E-3
Eutrophication terrestrial	EP-terrestrial	mol N eq.	6.33E-2	ND	ND	ND	ND	ND	ND	ND	ND	ND	3.24E-1	4.74E-2	1.01E-1	9.42E-2	2.11E-2
Photochemical ozone formation	POCP	kg NMVOC eq.	1.99E-2	ND	ND	ND	ND	ND	ND	ND	ND	ND	9.61E-2	1.83E-2	3.00E-2	3.28E-2	4.04E-3
Depletion of abiotic resources - minerals and metals	ADP-minerals&metals[1]	kg Sb eq.	5.82E-6	ND	ND	ND	ND	ND	ND	ND	ND	ND	2.49E-6	1.26E-5	7.76E-7	4.30E-6	-3.19E-6
Depletion of abiotic resources - fossil fuels	ADP-fossil[1]	MJ, net calorific value	4.51E+1	ND	ND	ND	ND	ND	ND	ND	ND	ND	9.16E+1	5.45E+1	2.86E+1	7.63E+1	-2.28E+1
Water use	WDP[1]	m^3 world eq. deprived	3.49E-1	ND	ND	ND	ND	ND	ND	ND	ND	ND	2.27E-1	2.56E-1	7.07E-2	2.37E-1	-1.79E-1
Acronyms	GWP-fossil = Global Warming Potential fossil fuels; GWP-biogenic = Global Warming Potential biogenic; GWP-luluc = Global Warming Potential land use and land use change; ODP = Depletion potential of the stratospheric ozone layer; AP = Acidification potential, Accumulated Exceedance; EP-freshwater = Eutrophication potential, fraction of nutrients reaching freshwater end compartment; EP-marine = Eutrophication potential, fraction of nutrients reaching marine end compartment; EP-terrestrial = Eutrophication potential, Accumulated Exceedance; POCP = Formation potential of tropospheric ozone; ADP-minerals&metals = Abiotic depletion potential for non-fossil resources; ADP-fossil = Abiotic depletion for fossil resources potential; WDP = Water (user) deprivation potential, deprivation-weighted water consumption																
General disclaimer	The results of the end-of-life stage (modules C1-C4) should be considered when using the results of the product stage (modules A1-A3/A1-A5 for services).																
Disclaimer 1	The results of this environmental impact indicator shall be used with care as the uncertainties of these results are high or as there is limited experience with the indicator																

Additional mandatory environmental performance indicators

Impact category	Indicator	Unit	A1-A3	A4	A5	B1	B2	B3	B4	B5	B6	B7	C1	C2	C3	C4	D
Climate change - GWP-GHG	GWP-GHG[1]	kg CO_2 eq.	2.57E+0	ND	ND	ND	ND	ND	ND	ND	ND	ND	6.95E+0	3.85E+0	2.17E+0	3.04E+0	-5.68E-1
Acronyms	GWP-GHG = Global warming potential greenhouse gas.																
General disclaimer	The results of the end-of-life stage (modules C1-C4) should be considered when using the results of the product stage (modules A1-A3/A1-A5 for services).																
Disclaimer 1	The GWP-GHG indicator is termed GWP-IOBC/GHG in the ILCD+EPD+ data format. The indicator accounts for all greenhouse gases except biogenic carbon dioxide uptake and emissions and biogenic carbon stored in the product. As such, the indicator is identical to GWP-total except that the CF for biogenic CO_2 is set to zero.																

Additional voluntary environmental performance indicators according to EN 15804

Impact category	Indicator	Unit	A1-A3	A4	A5	B1	B2	B3	B4	B5	B6	B7	C1	C2	C3	C4	D
Particulate matter emissions	PM	Disease incidence	3.15E-7	ND	ND	ND	ND	ND	ND	ND	ND	ND	1.80E-6	3.05E-7	4.28E-6	5.01E-7	3.86E-6
Ionizing radiation - human health	IRP[1]	kBq U235 eq.	2.88E-2	ND	ND	ND	ND	ND	ND	ND	ND	ND	4.32E-2	4.61E-2	1.35E-2	4.80E-2	-1.33E-2
Eco-toxicity - freshwater	ETP-fw[2]	CTUe	1.57E+1	ND	ND	ND	ND	ND	ND	ND	ND	ND	4.35E+1	3.02E+1	1.36E+1	3.55E+1	-6.33E+0
Human toxicity - cancer effects	HTP-c[2]	CTUh	7.74E-10	ND	ND	ND	ND	ND	ND	ND	ND	ND	2.13E-9	1.74E-9	6.65E-10	1.30E-9	-1.61E-10
Human toxicity - non-cancer effects	HTP-nc[2]	CTUh	1.32E-8	ND	ND	ND	ND	ND	ND	ND	ND	ND	1.49E-8	3.91E-8	4.66E-9	1.64E-8	-7.24E-9
Land-use related impacts/soil quality	SQP[2]	Dimensionless	2.21E+2	ND	ND	ND	ND	ND	ND	ND	ND	ND	6.11E+0	3.22E+1	1.91E+0	1.50E+2	-7.82E+1
Acronyms	PM = Potential incidence of disease due to particulate matter emissions; IRP = Potential human exposure efficiency relative to U235; ETP-fw = Potential comparative toxic unit for ecosystems; HTP-c = Potential comparative toxic unit for humans; HTP-nc = Potential comparative toxic unit for humans; SQP = Potential soil quality index.																
General disclaimer	The results of the end-of-life stage (modules C1-C4) should be considered when using the results of the product stage (modules A1-A3/A1-A5 for services).																
Disclaimer 1	This impact category deals mainly with the eventual impact of low dose ionizing radiation on human health of the nuclear fuel cycle. It does not consider effects due to possible nuclear accidents, occupational exposure nor due to radioactive waste disposal in underground facilities. Potential ionizing radiation from the soil, from radon and from some construction materials is also not measured by this indicator.																
Disclaimer 2	The results of this environmental impact indicator shall be used with care as the uncertainties of these results are high or as there is limited experience with the indicator.																

Resource use indicators according to EN 15804

Indicator	Unit	A1-A3	A4	A5	B1	B2	B3	B4	B5	B6	B7	C1	C2	C3	C4	D
PERE	MJ, net calorific value	4.18E+1	ND	ND	ND	ND	ND	ND	ND	ND	ND	5.18E-1	6.88E-1	1.62E-1	6.41E-1	-2.55E+0
PERM	MJ, net calorific value	3.32E+0	ND	ND	ND	ND	ND	ND	ND	ND	ND	0.00E+0	0.00E+0	0.00E+0	0.00E+0	0.00E+0
PERT	MJ, net calorific value	3.41E+0	ND	ND	ND	ND	ND	ND	ND	ND	ND	5.18E-1	6.88E-1	1.62E-1	6.41E-1	-2.55E+0
PENRE	MJ, net calorific value	4.18E+1	ND	ND	ND	ND	ND	ND	ND	ND	ND	8.32E+1	4.98E+1	2.60E+1	6.94E+1	-2.13E+1
PENRM	MJ, net calorific value	3.32E+0	ND	ND	ND	ND	ND	ND	ND	ND	ND	8.36E+0	4.66E+0	2.61E+0	6.88E+0	-1.53E+0
PENRT	MJ, net calorific value	4.51E+1	ND	ND	ND	ND	ND	ND	ND	ND	ND	9.16E+1	5.45E+1	2.86E+1	7.63E+1	-2.28E+1
SM	kg	1.55E-1	ND	ND	ND	ND	ND	ND	ND	ND	ND	5.29E-2	4.00E-2	1.65E-2	3.37E-2	-1.10E-1
RSF	MJ, net calorific value	8.37E-2	ND	ND	ND	ND	ND	ND	ND	ND	ND	5.83E-3	4.93E-3	1.82E-3	6.55E-3	-6.41E-2
NRSF	MJ, net calorific value	1.15E-2	ND	ND	ND	ND	ND	ND	ND	ND	ND	1.58E-2	2.26E-2	4.93E-3	1.68E-2	-5.14E-3
FW	m^3	8.03E-3	ND	ND	ND	ND	ND	ND	ND	ND	ND	4.90E-3	6.44E-3	1.53E-3	7.88E-2	-4.14E-3
Acronyms	PERE = Use of renewable primary energy excluding renewable primary energy resources used as raw materials; PERM = Use of renewable primary energy resources used as raw materials; PERT = Total use of renewable primary energy resources; PENRE = Use of non-renewable primary energy excluding non-renewable primary energy resources used as raw materials; PENRM = Use of non-renewable primary energy resources used as raw materials; PENRT = Total use of non-renewable primary energy re-sources; SM = Use of secondary material; RSF = Use of renewable secondary fuels; NRSF = Use of non-renewable secondary fuels; FW = Use of net fresh water.															
General disclaimer	The results of the end-of-life stage (modules C1-C4) should be considered when using the results of the product stage (modules A1-A3/A1-A5 for services).															

Waste indicators according to EN 15804

Indicator	Unit	A1-A3	A4	A5	B1	B2	B3	B4	B5	B6	B7	C1	C2	C3	C4	D
HWD	kg	3.02E-2	ND	ND	ND	ND	ND	ND	ND	ND	ND	7.61E-2	6.22E-2	2.38E-2	5.24E-2	-7.21E-3
NHWD	kg	5.65E-2	ND	ND	ND	ND	ND	ND	ND	ND	ND	5.64E-2	2.56E+0	1.76E-2	5.00E+2	-2.71E-2
RWD	kg	6.97E-6	ND	ND	ND	ND	ND	ND	ND	ND	ND	9.97E-6	1.10E-5	3.11E-6	1.12E-5	-3.24E-6
Acronyms	HWD = Hazardous waste disposed; NHWD = Non-hazardous waste disposed; RWD = Radioactive waste disposed.															
General disclaimer	The results of the end-of-life stage (modules C1-C4) should be considered when using the results of the product stage (modules A1-A3/A1-A5 for services).															

Output flow indicators according to EN 15804

Indicator	Unit	A1-A3	A4	A5	B1	B2	B3	B4	B5	B6	B7	C1	C2	C3	C4	D
CRU	kg	0.00E+0	ND	ND	ND	ND	ND	ND	ND	ND	ND	0.00E+0	0.00E+0	0.00E+0	0.00E+0	0.00E+0
MFR	kg	1.48E-1	ND	ND	ND	ND	ND	ND	ND	ND	ND	4.35E-2	3.52E-2	1.36E-2	2.76E-2	-1.06E-1
MER	kg	0.00E+0	ND	ND	ND	ND	ND	ND	ND	ND	ND	0.00E+0	0.00E+0	0.00E+0	0.00E+0	0.00E+0
EEE	MJ, net calorific value	0.00E+0	ND	ND	ND	ND	ND	ND	ND	ND	ND	0.00E+0	0.00E+0	0.00E+0	0.00E+0	0.00E+0
EET	MJ, net calorific value	0.00E+0	ND	ND	ND	ND	ND	ND	ND	ND	ND	0.00E+0	0.00E+0	0.00E+0	0.00E+0	0.00E+0
Acronyms	CRU = Components for re-use; MFR = Materials for recycling; MER = Materials for energy recovery; EEE = Exported electrical energy; EET = Exported thermal energy.															
General disclaimer	The results of the end-of-life stage (modules C1-C4) should be considered when using the results of the product stage (modules A1-A3/A1-A5 for services).															

Results for additional scenarios for modules A4-C4

Additional scenario	Alternative EOL Full Reuse Scenario
Description of the scenario/method	-Material Recovery Rate Main scenario: 50% of the product is recycled, 50% is landfilled. Additional scenario: 100% is reused as secondary aggregate. -End-of-Life Treatment Main scenario: Includes impacts from landfilling and partial recycling. Additional scenario: Avoids disposal impacts entirely Additional scenario represents an idealized best-case situation, reflecting a fully optimized circular economy approach, and does not correspond to current average waste management practices

Impact category	Indicator	Unit	A1-A3	A4	A5	B1	B2	B3	B4	B5	B6	B7	C1	C2	C3	C4	D
Climate change - total	GWP-total	kg CO2 eq.	2.57E+0	ND	ND	ND	ND	ND	ND	ND	ND	ND	6.95E+0	3.85E+0	4.34E+0	0.00E+0	-1.14E+0
Climate change - fossil	GWP-fossil	kg CO2 eq.	2.57E+0	ND	ND	ND	ND	ND	ND	ND	ND	ND	6.95E+0	3.85E+0	4.34E+0	0.00E+0	-1.14E+0
Climate change - biogenic	GWP-biogenic	kg CO2 eq.	2.56E-3	ND	ND	ND	ND	ND	ND	ND	ND	ND	8.71E-4	9.98E-4	5.53E-4	0.00E+0	-3.63E-3
Climate change - land use and land-use change	GWP-luluc	kg CO2 eq.	3.80E-4	ND	ND	ND	ND	ND	ND	ND	ND	ND	7.81E-4	1.98E-3	4.88E-4	0.00E+0	2.73E-4
Ozone depletion	ODP	kg CFC-11 eq.	9.13E-8	ND	ND	ND	ND	ND	ND	ND	ND	ND	1.10E-7	5.75E-8	6.90E-8	0.00E+0	-1.18E-7
Acidification	AP	mol H+ eq.	1.63E-2	ND	ND	ND	ND	ND	ND	ND	ND	ND	6.44E-2	1.36E-2	4.02E-2	0.00E+0	2.01E-3
Eutrophication aquatic freshwater	EP-freshwater	kg P eq.	6.66E-4	ND	ND	ND	ND	ND	ND	ND	ND	ND	2.13E-4	3.12E-4	1.33E-4	0.00E+0	-9.44E-4
Eutrophication aquatic marine	EP-marine	kg N eq.	5.60E-3	ND	ND	ND	ND	ND	ND	ND	ND	ND	2.98E-2	4.47E-3	1.86E-2	0.00E+0	3.97E-3
Eutrophication terrestrial	EP-terrestrial	mol N eq.	6.33E-2	ND	ND	ND	ND	ND	ND	ND	ND	ND	3.24E-1	4.74E-2	2.03E-1	0.00E+0	4.22E-2
Photochemical ozone formation	POCP	kg NMVOC eq.	1.99E-2	ND	ND	ND	ND	ND	ND	ND	ND	ND	9.61E-2	1.83E-2	6.00E-2	0.00E+0	8.08E-3
Depletion of abiotic resources - minerals and metals	ADP-minerals & metals	kg Sb eq.	5.82E-6	ND	ND	ND	ND	ND	ND	ND	ND	ND	2.49E-6	1.26E-5	1.55E-6	0.00E+0	-6.38E-6
Depletion of abiotic resources - fossil fuels	ADP-fossil	MJ, net calorific value	4.51E+1	ND	ND	ND	ND	ND	ND	ND	ND	ND	6.95E+0	3.85E+0	4.34E+0	0.00E+0	-1.14E+0
Water use	WDP	m3 world eq. deprived	3.49E-1	ND	ND	ND	ND	ND	ND	ND	ND	ND	2.27E-1	2.56E-1	1.41E-1	0.00E+0	-3.57E-1
Climate change - GWP-GHG	GWP-GHG	kg CO2 eq.	2.57E+0	ND	ND	ND	ND	ND	ND	ND	ND	ND	6.95E+0	3.85E+0	4.34E+0	0.00E+0	-1.14E+0
Particulate matter emissions	PM	Disease incidence	3.15E-7	ND	ND	ND	ND	ND	ND	ND	ND	ND	1.80E-6	3.05E-7	8.56E-6	0.00E+0	7.71E-6
Ionizing radiation - human health	IRP	kBq U235 eq.	2.88E-2	ND	ND	ND	ND	ND	ND	ND	ND	ND	4.32E-2	4.61E-2	2.69E-2	0.00E+0	-2.66E-2
Eco-toxicity - freshwater	ETP-fw	CTUe	1.57E+1	ND	ND	ND	ND	ND	ND	ND	ND	ND	4.35E+1	3.02E+1	2.71E+1	0.00E+0	-1.27E+1

Human toxicity - cancer effects	HTP-c	CTUh	7.74E-10	ND	ND	ND	ND	ND	ND	ND	ND	ND	2.13E-9	1.74E-9	1.33E-9	0.00E+0	-3.23E-10
Human toxicity - non-cancer effects	HTP-nc	CTUh	1.32E-8	ND	ND	ND	ND	ND	ND	ND	ND	ND	1.49E-8	3.91E-8	9.33E-9	0.00E+0	-1.45E-8
Land-use related impacts/soil quality	SQP	Dimensionless	2.21E+2	ND	ND	ND	ND	ND	ND	ND	ND	ND	6.11E+0	3.22E+1	3.81E+0	0.00E+0	-1.56E+2
Use of renewable primary energy excluding renewable primary energy resources used as raw materials	PERE	MJ, net calorific value	4.18E+1	ND	ND	ND	ND	ND	ND	ND	ND	ND	5.18E-1	6.88E-1	3.23E-1	0.00E+0	-5.10E+0
Use of renewable primary energy resources used as raw materials	PERM	MJ, net calorific value	3.32E+0	ND	ND	ND	ND	ND	ND	ND	ND	ND	0.00E+0	0.00E+0	0.00E+0	0.00E+0	0.00E+0
Total use of renewable primary energy resources	PERT	MJ, net calorific value	3.41E+0	ND	ND	ND	ND	ND	ND	ND	ND	ND	5.18E-1	6.88E-1	3.23E-1	0.00E+0	-5.10E+0
Use of non-renewable primary energy excluding non-renewable primary energy resources used as raw materials	PENRE	MJ, net calorific value	4.18E+1	ND	ND	ND	ND	ND	ND	ND	ND	ND	8.32E+1	4.98E+1	5.20E+1	0.00E+0	-4.26E+1
Use of non-renewable primary energy resources used as raw materials	PENRM	MJ, net calorific value	3.32E+0	ND	ND	ND	ND	ND	ND	ND	ND	ND	8.36E+0	4.66E+0	5.22E+0	0.00E+0	-3.06E+0
Total use of non-renewable primary energy re-sources	PENRT	MJ, net calorific value	4.15E+1	ND	ND	ND	ND	ND	ND	ND	ND	ND	9.16E+1	5.45E+1	5.72E+1	0.00E+0	-4.57E+1
Use of secondary material	SM	kg	1.55E-1	ND	ND	ND	ND	ND	ND	ND	ND	ND	5.29E-2	4.00E-2	3.30E-2	0.00E+0	-2.19E-1
Use of renewable secondary fuels	RSF	MJ, net calorific value	8.37E-2	ND	ND	ND	ND	ND	ND	ND	ND	ND	5.83E-3	4.93E-3	3.64E-3	0.00E+0	-1.28E-1
Use of non-renewable secondary fuels	NRSF	MJ, net calorific value	1.15E-2	ND	ND	ND	ND	ND	ND	ND	ND	ND	1.58E-2	2.26E-2	9.85E-3	0.00E+0	-1.03E-2
Use of net fresh water	FW	m3	8.03E-3	ND	ND	ND	ND	ND	ND	ND	ND	ND	4.90E-3	6.44E-3	3.06E-3	0.00E+0	-8.28E-3
Hazardous waste disposed	HWD	kg	3.02E-2	ND	ND	ND	ND	ND	ND	ND	ND	ND	7.61E-2	6.22E-2	4.75E-2	0.00E+0	-1.44E-2
Non-hazardous waste disposed	NHWD	kg	5.65E-2	ND	ND	ND	ND	ND	ND	ND	ND	ND	5.64E-2	2.56E+0	3.52E-2	0.00E+0	-5.42E-2
Radioactive waste disposed	RWD	kg	6.97E-6	ND	ND	ND	ND	ND	ND	ND	ND	ND	9.97E-6	1.10E-5	6.22E-6	0.00E+0	-6.48E-6
Components for re-use	CRU	kg	0.00E+0	ND	ND	ND	ND	ND	ND	ND	ND	ND	0.00E+0	0.00E+0	0.00E+0	0.00E+0	0.00E+0
Materials for recycling	MFR	kg	1.48E-1	ND	ND	ND	ND	ND	ND	ND	ND	ND	4.35E-2	3.52E-2	2.72E-2	0.00E+0	-2.13E-1
Materials for energy recovery	MER	kg	0.00E+0	ND	ND	ND	ND	ND	ND	ND	ND	ND	0.00E+0	0.00E+0	0.00E+0	0.00E+0	0.00E+0
Exported electrical energy	EEE	MJ, net calorific value	0.00E+0	ND	ND	ND	ND	ND	ND	ND	ND	ND	0.00E+0	0.00E+0	0.00E+0	0.00E+0	0.00E+0
Exported thermal energy	EET	MJ, net calorific value	0.00E+0	ND	ND	ND	ND	ND	ND	ND	ND	ND	0.00E+0	0.00E+0	0.00E+0	0.00E+0	0.00E+0
Acronyms																	
Disclaimers	The same disclaimers apply as in the main scenario. These include mandatory, additional, and voluntary indicators as defined in EN 15804 and are reported in the "Environmental Performance" section of the EPD.																
General disclaimer	The results of the end-of-life stage (modules C1-C4) should be considered when using the results of the product stage (modules A1-A3/A1-A5 for services).																

Results for additional scenarios for modules A4-C4

Additional scenario	Alternative EOL Full Landfilling Scenario
Description of the scenario/method	-Material Recovery Rate: Main scenario: 50% of the product is recycled, 50% is landfilled. Alternative scenario: 100% of the product is landfilled, with no material recovery or reuse. -End-of-Life Treatment: Main scenario: Includes impacts from landfilling and partial recycling. Alternative scenario: Reflects complete disposal, with associated impacts from landfill processes. This scenario represents a worst-case situation, assuming no material is recovered at end-of-life.

Impact category	Indicator	Unit	A1-A3	A4	A5	B1	B2	B3	B4	B5	B6	B7	C1	C2	C3	C4	D
Climate change - total	GWP-total	kg CO2 eq.	2.57E+0	ND	ND	ND	ND	ND	ND	ND	ND	ND	6.95E+0	3.85E+0	0.00E+0	6.08E+0	0.00E+0
Climate change - fossil	GWP-fossil	kg CO2 eq.	2.57E+0	ND	ND	ND	ND	ND	ND	ND	ND	ND	6.95E+0	3.85E+0	0.00E+0	6.08E+0	0.00E+0
Climate change - biogenic	GWP-biogenic	kg CO2 eq.	2.56E-3	ND	ND	ND	ND	ND	ND	ND	ND	ND	8.71E-4	9.98E-4	0.00E+0	2.40E-3	0.00E+0
Climate change - land use and land-use change	GWP-luluc	kg CO2 eq.	3.80E-4	ND	ND	ND	ND	ND	ND	ND	ND	ND	7.81E-4	1.98E-3	0.00E+0	3.67E-3	0.00E+0
Ozone depletion	ODP	kg CFC-11 eq.	9.13E-8	ND	ND	ND	ND	ND	ND	ND	ND	ND	1.10E-7	5.75E-8	0.00E+0	1.76E-7	0.00E+0
Acidification	AP	mol H+ eq.	1.63E-2	ND	ND	ND	ND	ND	ND	ND	ND	ND	6.44E-2	1.36E-2	0.00E+0	4.58E-2	0.00E+0
Eutrophication aquatic freshwater	EP-freshwater	kg P eq.	6.66E-4	ND	ND	ND	ND	ND	ND	ND	ND	ND	2.13E-4	3.12E-4	0.00E+0	5.06E-4	0.00E+0
Eutrophication aquatic marine	EP-marine	kg N eq.	5.60E-3	ND	ND	ND	ND	ND	ND	ND	ND	ND	2.98E-2	4.47E-3	0.00E+0	1.76E-2	0.00E+0
Eutrophication terrestrial	EP-terrestrial	mol N eq.	6.33E-2	ND	ND	ND	ND	ND	ND	ND	ND	ND	3.24E-1	4.74E-2	0.00E+0	1.88E-1	0.00E+0
Photochemical ozone formation	POCP	kg NMVOC eq.	1.99E-2	ND	ND	ND	ND	ND	ND	ND	ND	ND	9.61E-2	1.83E-2	0.00E+0	6.56E-2	0.00E+0
Depletion of abiotic resources - minerals and metals	ADP-minerals & metals	kg Sb eq.	5.82E-6	ND	ND	ND	ND	ND	ND	ND	ND	ND	2.49E-6	1.26E-5	0.00E+0	8.61E-6	0.00E+0
Depletion of abiotic resources - fossil fuels	ADP-fossil	MJ, net calorific value	4.51E+1	ND	ND	ND	ND	ND	ND	ND	ND	ND	9.16E+1	5.45E+1	0.00E+0	1.53E+2	0.00E+0
Water use	WDP	m3 world eq. deprived	3.49E-1	ND	ND	ND	ND	ND	ND	ND	ND	ND	2.27E-1	2.56E-1	0.00E+0	4.74E-1	0.00E+0
Climate change - GWP-GHG	GWP-GHG	kg CO2 eq.	2.57E+0	ND	ND	ND	ND	ND	ND	ND	ND	ND	6.95E+0	3.85E+0	0.00E+0	6.08E+0	0.00E+0
Particulate matter emissions	PM	Disease incidence	3.15E-7	ND	ND	ND	ND	ND	ND	ND	ND	ND	1.80E-6	3.05E-7	0.00E+0	1.00E-6	0.00E+0
Ionizing radiation - human health	IRP	kBq U235 eq.	2.88E-2	ND	ND	ND	ND	ND	ND	ND	ND	ND	4.32E-2	4.61E-2	0.00E+0	9.59E-2	0.00E+0
Eco-toxicity - freshwater	ETP-fw	CTUe	1.57E+1	ND	ND	ND	ND	ND	ND	ND	ND	ND	4.35E+1	3.02E+1	0.00E+0	7.11E+1	0.00E+0
Human toxicity - cancer effects	HTP-c	CTUh	7.74E-10	ND	ND	ND	ND	ND	ND	ND	ND	ND	2.13E-9	1.74E-9	0.00E+0	2.60E-9	0.00E+0
Human toxicity - non-cancer effects	HTP-nc	CTUh	1.32E-8	ND	ND	ND	ND	ND	ND	ND	ND	ND	1.49E-8	3.91E-8	0.00E+0	3.27E-8	0.00E+0
Land-use related impacts/soil quality	SQP	Dimensionless	2.21E+2	ND	ND	ND	ND	ND	ND	ND	ND	ND	6.11E+0	3.22E+1	0.00E+0	3.01E+2	0.00E+0

Use of renewable primary energy excluding renewable primary energy resources used as raw materials	PERE	MJ, net calorific value	4.18E+1	ND	ND	ND	ND	ND	ND	ND	ND	ND	5.18E-1	6.88E-1	0.00E+0	1.28E+0	0.00E+0
Use of renewable primary energy resources used as raw materials	PERM	MJ, net calorific value	3.32E+0	ND	ND	ND	ND	ND	ND	ND	ND	ND	0.00E+0	0.00E+0	0.00E+0	0.00E+0	0.00E+0
Total use of renewable primary energy resources	PERT	MJ, net calorific value	3.41E+0	ND	ND	ND	ND	ND	ND	ND	ND	ND	5.18E-1	6.88E-1	0.00E+0	1.28E+0	0.00E+0
Use of non-renewable primary energy excluding non-renewable primary energy resources used as raw materials	PENRE	MJ, net calorific value	4.18E+1	ND	ND	ND	ND	ND	ND	ND	ND	ND	8.32E+1	4.98E+1	0.00E+0	1.39E+2	0.00E+0
Use of non-renewable primary energy resources used as raw materials	PENRM	MJ, net calorific value	3.32E+0	ND	ND	ND	ND	ND	ND	ND	ND	ND	8.36E+0	4.66E+0	0.00E+0	1.38E+1	0.00E+0
Total use of non-renewable primary energy re-sources	PENRT	MJ, net calorific value	4.51E+1	ND	ND	ND	ND	ND	ND	ND	ND	ND	9.16E+1	5.45E+1	0.00E+0	1.53E+2	0.00E+0
Use of secondary material	SM	kg	1.55E-1	ND	ND	ND	ND	ND	ND	ND	ND	ND	5.29E-2	4.00E-2	0.00E+0	6.73E-2	0.00E+0
Use of renewable secondary fuels	RSF	MJ, net calorific value	8.37E-2	ND	ND	ND	ND	ND	ND	ND	ND	ND	5.83E-3	4.93E-3	0.00E+0	1.31E-2	0.00E+0
Use of non-renewable secondary fuels	NRSF	MJ, net calorific value	1.15E-2	ND	ND	ND	ND	ND	ND	ND	ND	ND	1.58E-2	2.26E-2	0.00E+0	3.36E-2	0.00E+0
Use of net fresh water	FW	m3	8.03E-3	ND	ND	ND	ND	ND	ND	ND	ND	ND	4.90E-3	6.44E-3	0.00E+0	1.58E-1	0.00E+0
Hazardous waste disposed	HWD	kg	3.02E-2	ND	ND	ND	ND	ND	ND	ND	ND	ND	7.61E-2	6.22E-2	0.00E+0	1.05E-1	0.00E+0
Non-hazardous waste disposed	NHWD	kg	5.65E-2	ND	ND	ND	ND	ND	ND	ND	ND	ND	5.64E-2	2.56E+0	0.00E+0	1.00E+3	0.00E+0
Radioactive waste disposed.	RWD	kg	6.97E-6	ND	ND	ND	ND	ND	ND	ND	ND	ND	9.97E-6	1.10E-5	0.00E+0	2.24E-5	0.00E+0
Components for re-use	CRU	kg	0.00E+0	ND	ND	ND	ND	ND	ND	ND	ND	ND	0.00E+0	0.00E+0	0.00E+0	0.00E+0	0.00E+0
Materials for recycling	MFR	kg	1.48E-1	ND	ND	ND	ND	ND	ND	ND	ND	ND	4.35E-2	3.52E-2	0.00E+0	5.52E-2	0.00E+0
Materials for energy recovery	MER	kg	0.00E+0	ND	ND	ND	ND	ND	ND	ND	ND	ND	0.00E+0	0.00E+0	0.00E+0	0.00E+0	0.00E+0
Exported electrical energy	EEE	MJ, net calorific value	0.00E+0	ND	ND	ND	ND	ND	ND	ND	ND	ND	0.00E+0	0.00E+0	0.00E+0	0.00E+0	0.00E+0
Exported thermal energy	EET	MJ, net calorific value	0.00E+0	ND	ND	ND	ND	ND	ND	ND	ND	ND	0.00E+0	0.00E+0	0.00E+0	0.00E+0	0.00E+0
Acronyms																	
Disclaimers	The same disclaimers apply as in the main scenario. These include mandatory, additional, and voluntary indicators as defined in EN 15804 and are reported in the "Environmental Performance" section of the EPD.																
General disclaimer	The results of the end-of-life stage (modules C1-C4) should be considered when using the results of the product stage (modules A1-A3/A1-A5 for services).																

INFORMATION RELATED TO EPDS OF MULTIPLE PRODUCTS

Description of how the averages have been determined	The results presented in this EPD refer to three similar products manufactured at the same production site. As the environmental performance across the products was comparable, an average was derived using site-specific data. This average is considered to reflect the typical impacts of the product group as currently produced.

ABBREVIATIONS

Not applicable

REFERENCES

General Programme Instructions of the International EPD® System. Version 5.0.1

PCR 2019:14 v.2.0.1 Construction products. EPD System.

EN 15804:2012+A2:2019/AC:2021, Sustainability of construction works — Environmental Product Declarations - Core rules for the product category of construction products

ISO 14020:2000 Environmental labels and declarations - General principles

ISO 14025:2006 Environmental labels and declarations - Type III environmental declarations - Principles and procedures

ISO 14040:2006 Environmental management - Life cycle assessment - Principles and framework

ISO 14044:2006 Environmental management — Life cycle assessment — Requirements and guidelines

Residual Energy Mix 2024 from Renewable Energy Sources Operator & Guarantees of Origin (DAPEEP SA)

VERSION HISTORY

2025-10-06/ EPD Version 001: Original version of EPD (unpublished version)

2025-10-09/ EPD Version 002: Added additional scenarios, added product photos, changes made to product data (unpublished version)

2025-10-13/ EPD Version 003: Added additional scenarios results, changes made to some indicator results

2025-12-23/ EPD Version 004: Addition of crushed aggregate 0/22.4 (3A).